AIDD8122 — Advanced Topics in Artificial Intelligence & Drug Discovery

Molecular representations, model families and evaluation protocols for predicting BACE-1 inhibition

A systematic benchmark against graph neural networks

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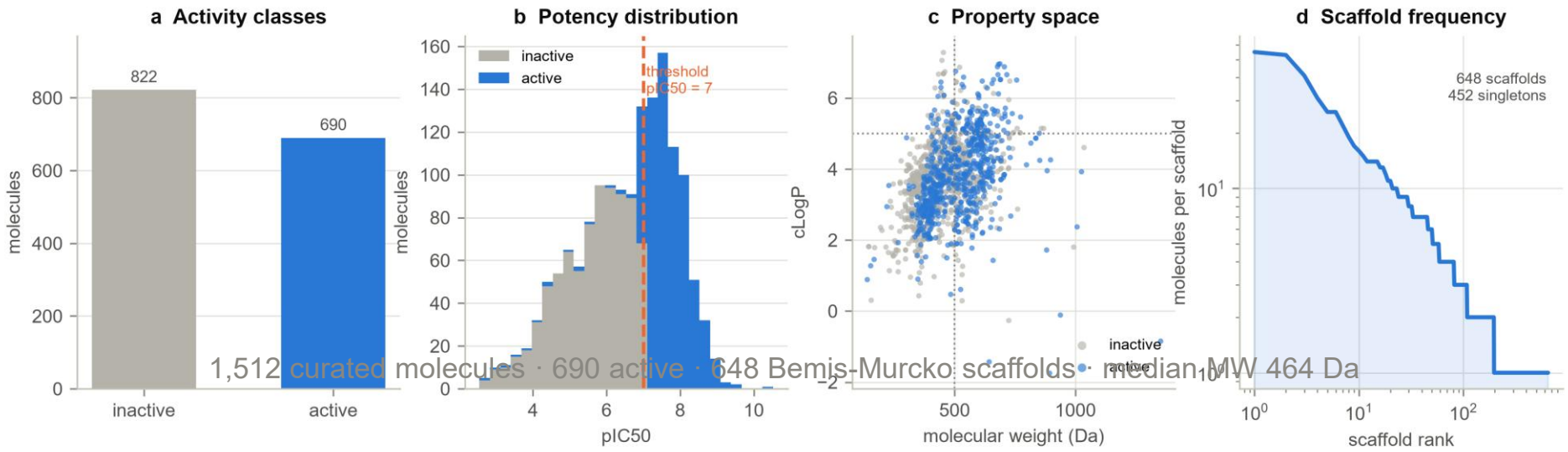
MOTIVATION

BACE-1: a well-studied target that keeps failing

— Alzheimer's disease: 55 million people, no reliable disease-modifying therapy. BACE-1 performs the first, rate-limiting cut that releases amyloid-beta.

— Four clinical candidates reached late-stage trials; all were discontinued — but two decades of chemistry left a rich, well-annotated dataset.

→ An ide



RESEARCH QUESTIONS

Three choices decide what a model appears to be worth

Every QSAR study makes these choices — usually implicitly. This project makes them the object of study.

RQ1

Representation vs model

Which choice buys more accuracy for the same effort?

RQ2

Random vs scaffold split

How optimistic is a random split, in numbers?

RQ3

Learned vs fixed features

Does a graph network beat fingerprints at this scale?

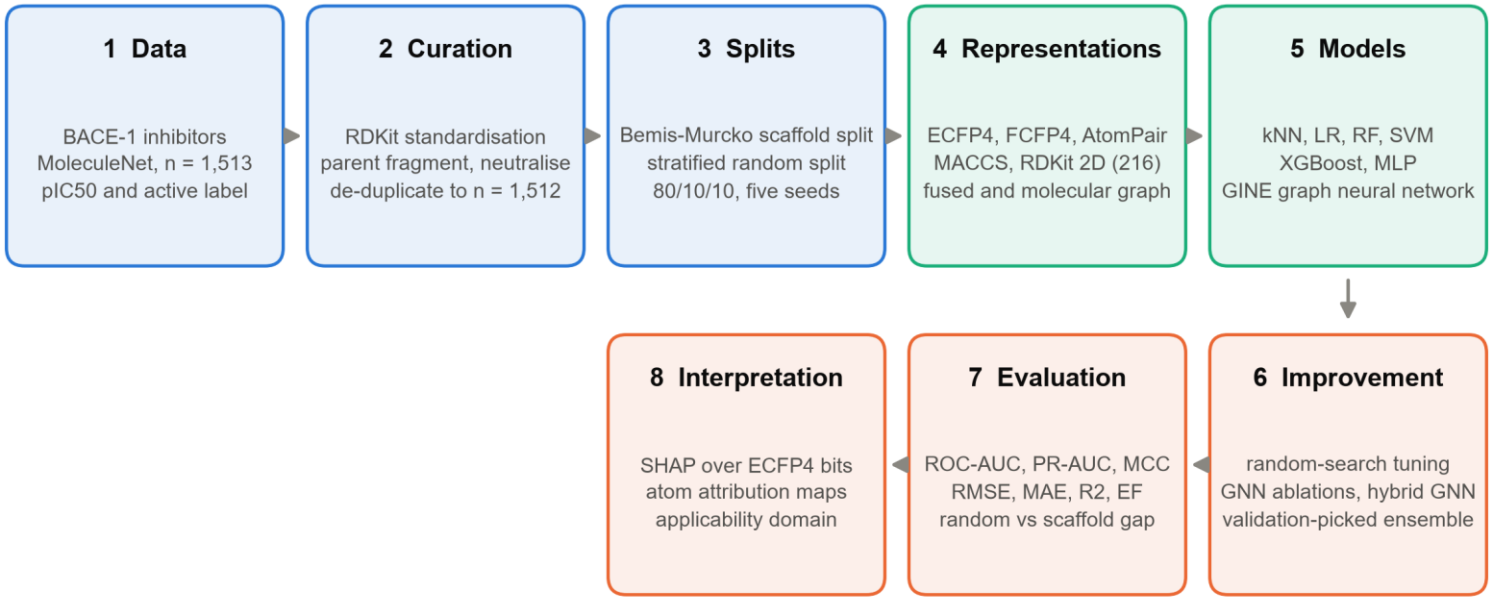
RQ4

Accuracy vs reliability

Can we say where the model should be trusted?

METHODS

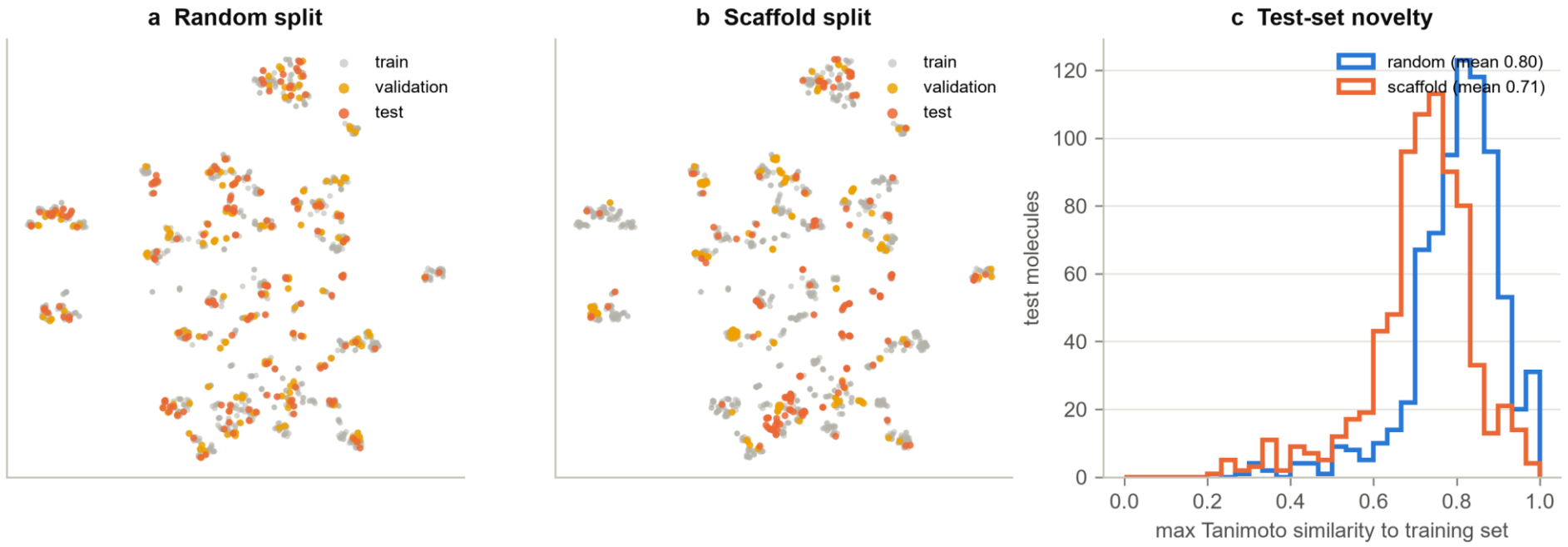
One dataset, one protocol, everything else varied



- 6 representations × 6 model families × 2 tasks × 2 splits × 5 seeds, plus a GINE graph network written from scratch in PyTorch.
- Every selection decision — hyperparameters, early stopping, ensemble membership — made on validation data only.

EVALUATION DESIGN

Scaffold splitting genuinely hides chemistry from the model

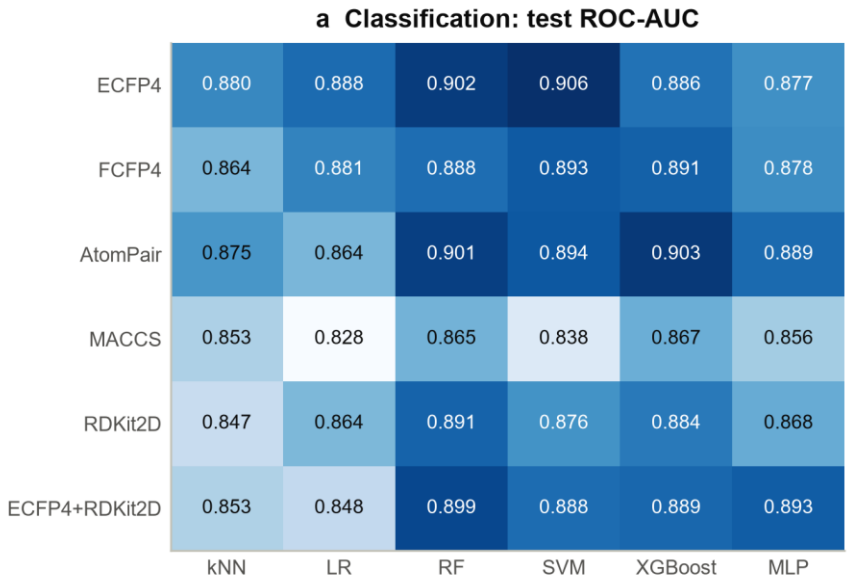


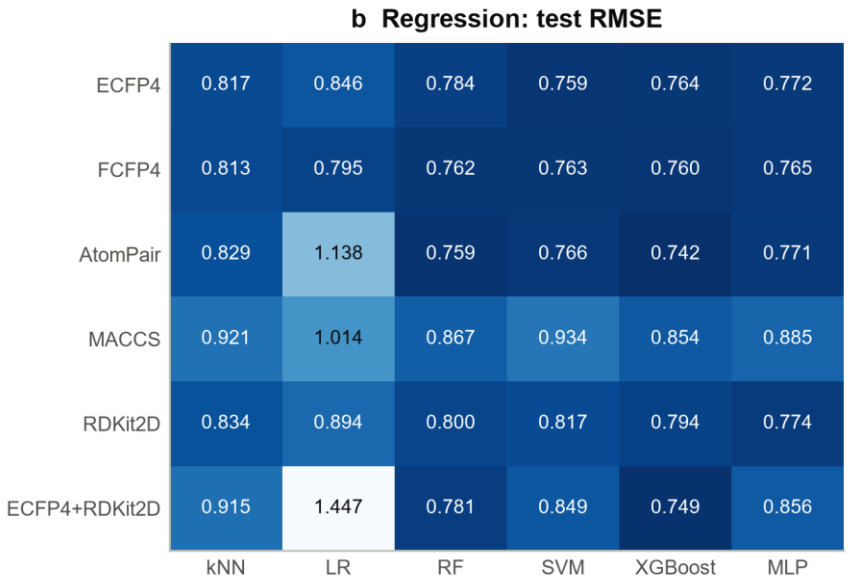
0.80
mean max Tanimoto similarity, random split

0.71
mean max Tanimoto similarity, scaffold split

648
distinct Bemis-Murcko scaffolds

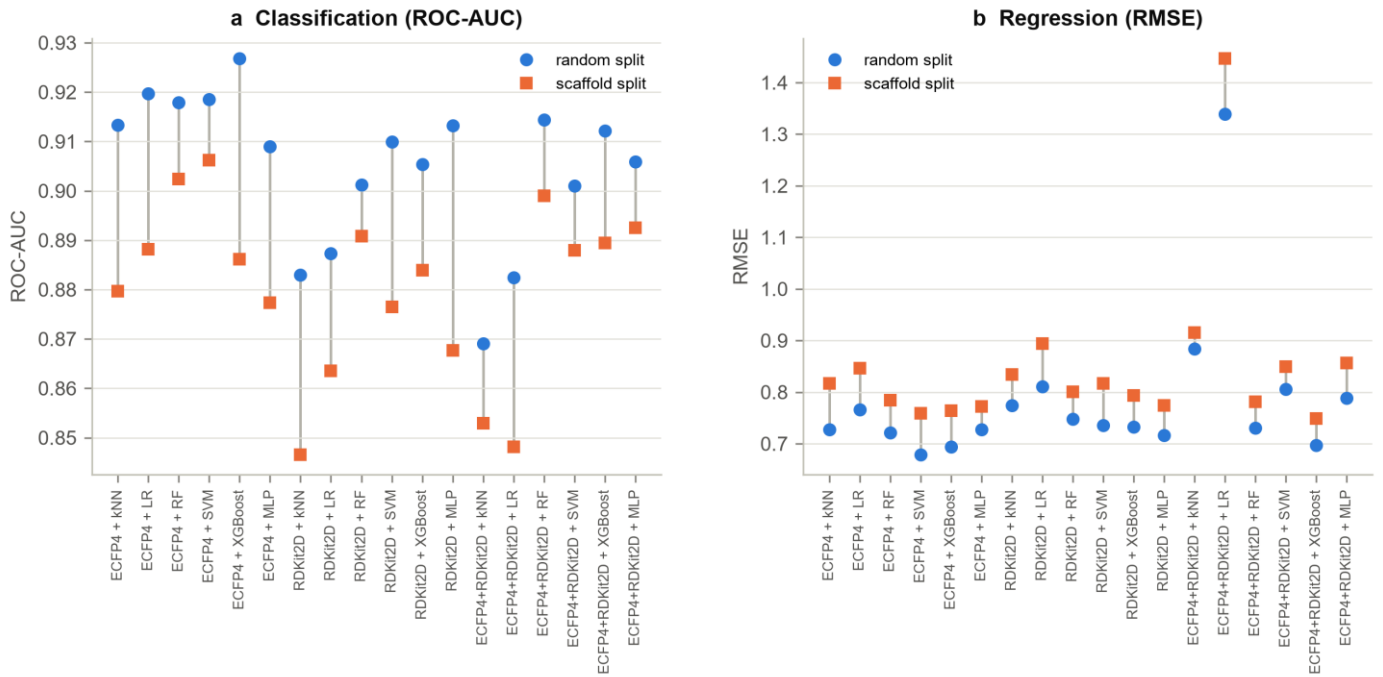
Representation matters at least as much as the model





- Spread between best and worst representation: 0.039 ROC-AUC. Between best and worst model family: 0.029.
- Best classifier ECFP4 + SVM (0.906); best regressor AtomPair + XGBoost (0.742 RMSE).
- Naive fusion of fingerprints and descriptors did not help — it hurt the linear model.

A random split flatters every single model

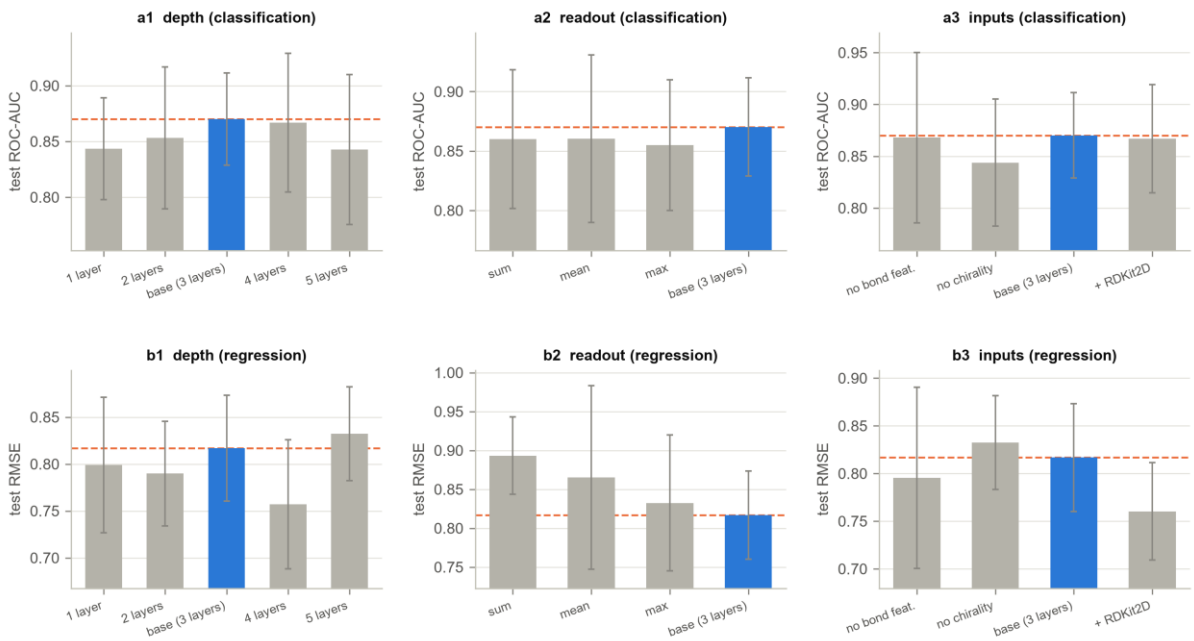


18/18
configurations scored better under random splitting

+0.025
mean ROC-AUC inflation

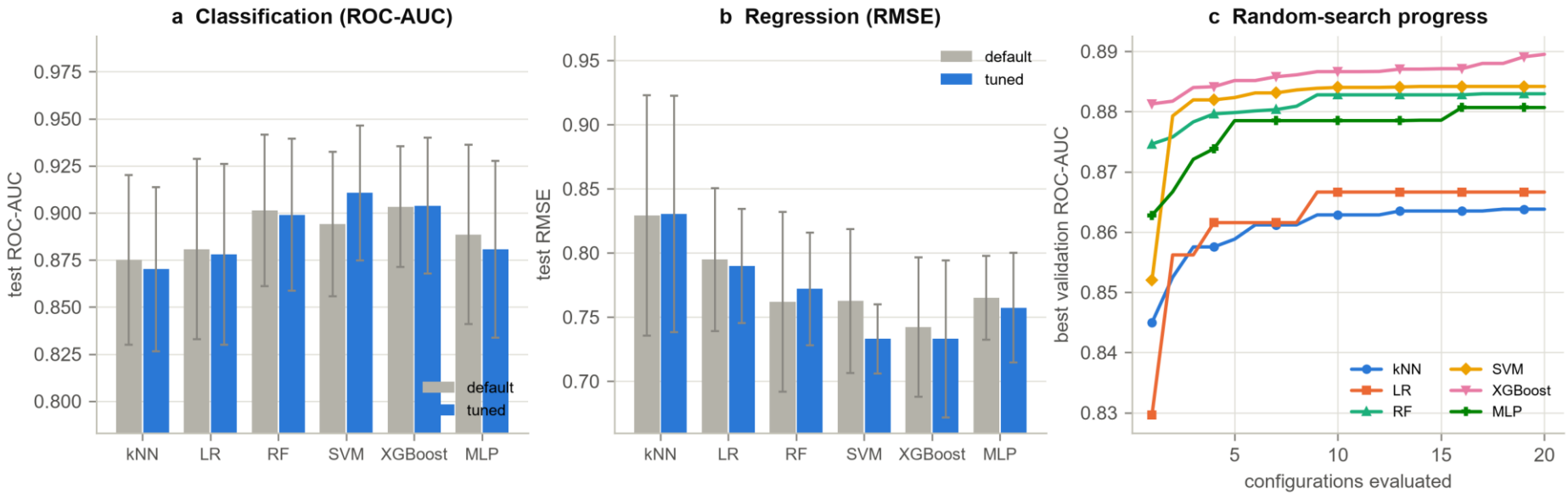
−0.065
mean RMSE understatement (log units)

The graph network does not win at 1,200 molecules



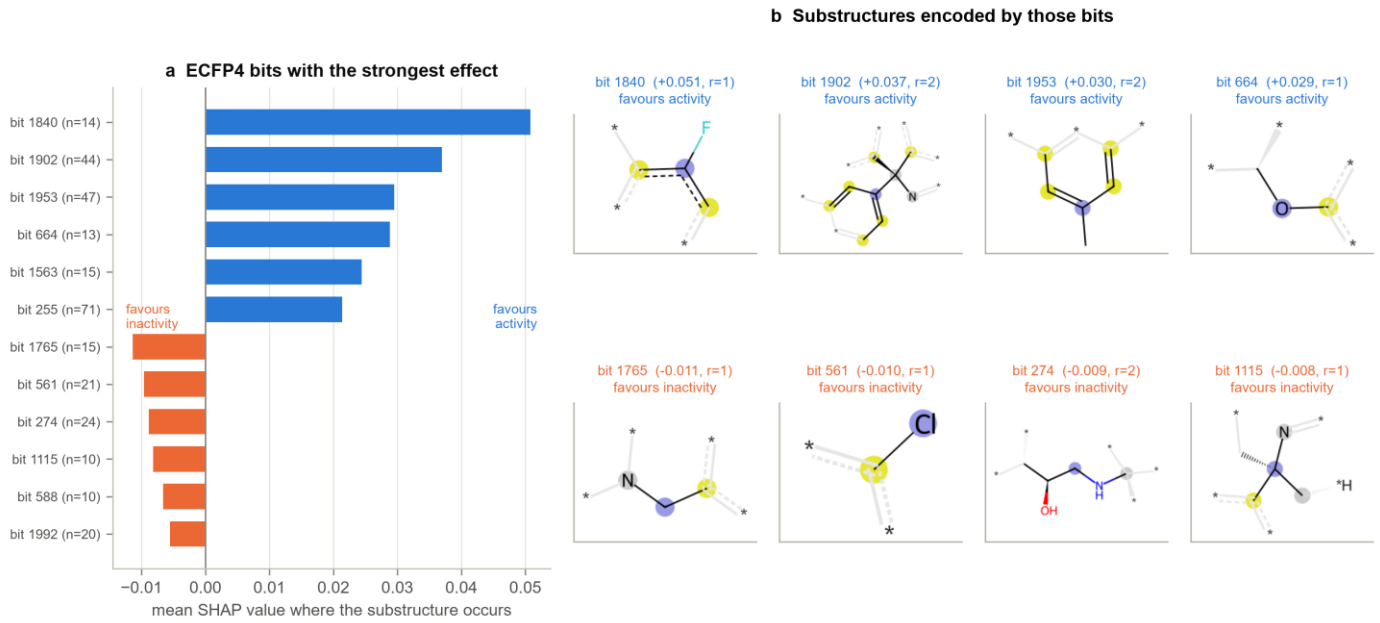
- GNN reaches 0.870 ROC-AUC and 0.817 RMSE — behind the best fingerprint models (0.906 / 0.742).
- Ablations behave as theory predicts: depth peaks at three layers (over-smoothing beyond), concatenated readout beats any single pooling.
- Removing stereochemistry costs 0.026 ROC-AUC — chemically sensible for transition-state isosteres.

Tuning buys little; ensembling helps only regression



- 20 random configurations per model, task and seed: mean change -0.0001 ROC-AUC, only 2 of 6 families improved.
- With ~150 validation molecules, the best of twenty captures noise as readily as signal: validation keeps rising, test does not follow.
- Ensembling: no gain for classification, but the best overall regression result (0.707 RMSE).

The model's reasoning is chemically legible



- SHAP over ECFP4 bits maps each contribution back to a concrete substructure: fluorinated and aromatic-nitrogen environments raise predicted activity.
- Presence of a substructure is informative; absence is not — a direct consequence of how fingerprints encode molecules.

A single accuracy number hides a twofold difference in reliability



67% → 88%

accuracy, least to most similar to the training set

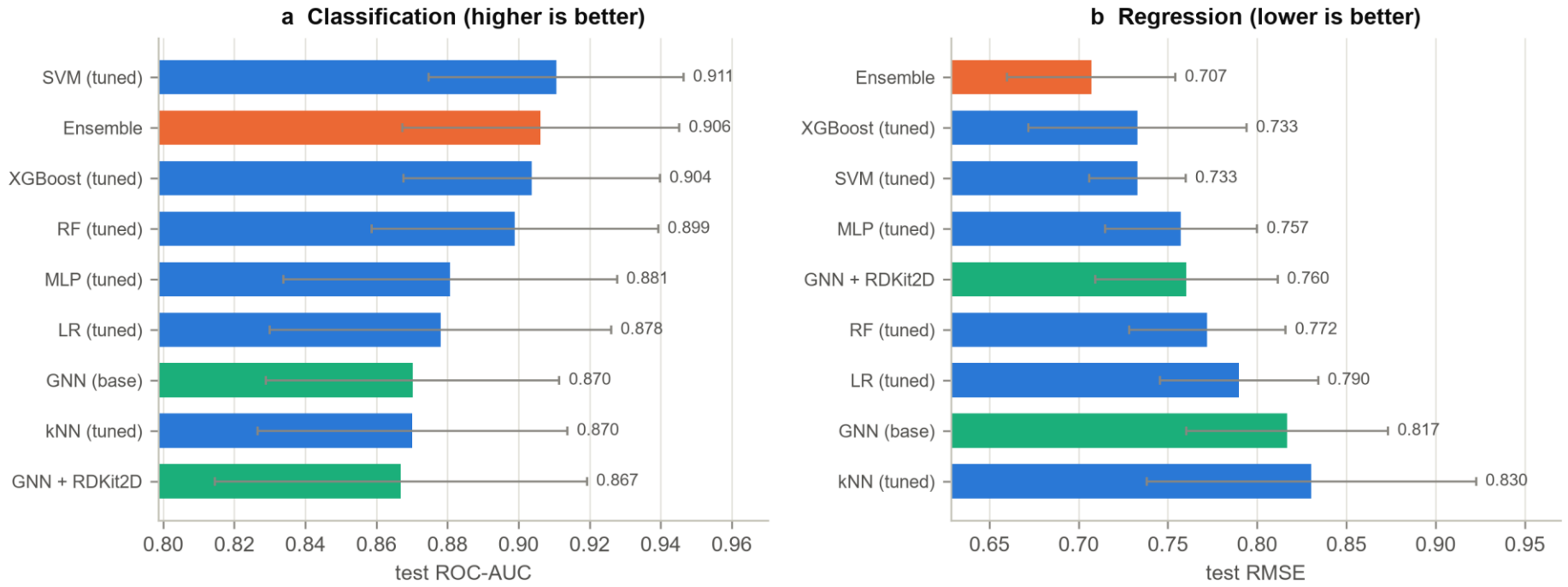
1.41 → 0.60

pIC50 RMSE over the same range

97%

accuracy on the most confident 10%

Everything side by side



Scaffold-split test performance, mean ± SD over five seeds. Blue: classical models. Green: graph networks. Orange: ensemble.

What to take away

1 Spend effort on the representation before the model — it explains as much or more of the variation.

2 Report scaffold-split numbers. A random split inflated every configuration here, by 0.025 ROC-AUC on average.

3 At ~1,200 molecules a learned graph representation does not beat well-designed fingerprints; descriptors recover part of its gap.

4 Tuning and ensembling give little on a small validation set — the noise floor is reached quickly.

5 Always publish an applicability domain: reliability varied twofold with similarity to the training set, and confidence tracked it closely enough to triage.

Thank you

Questions welcome

Code, cached splits, per-molecule predictions and figure scripts:

https://github.com/ljyhh0826-coder/AIDD8122-BACE1-ML